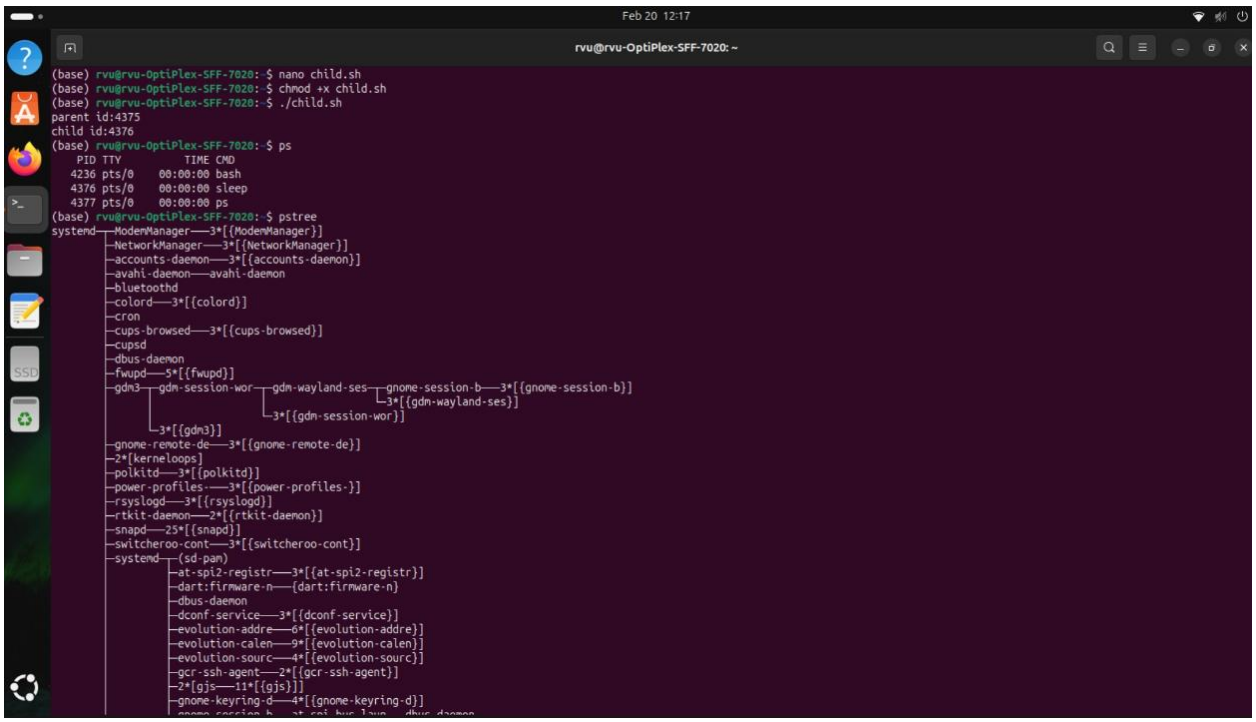
OS LAB

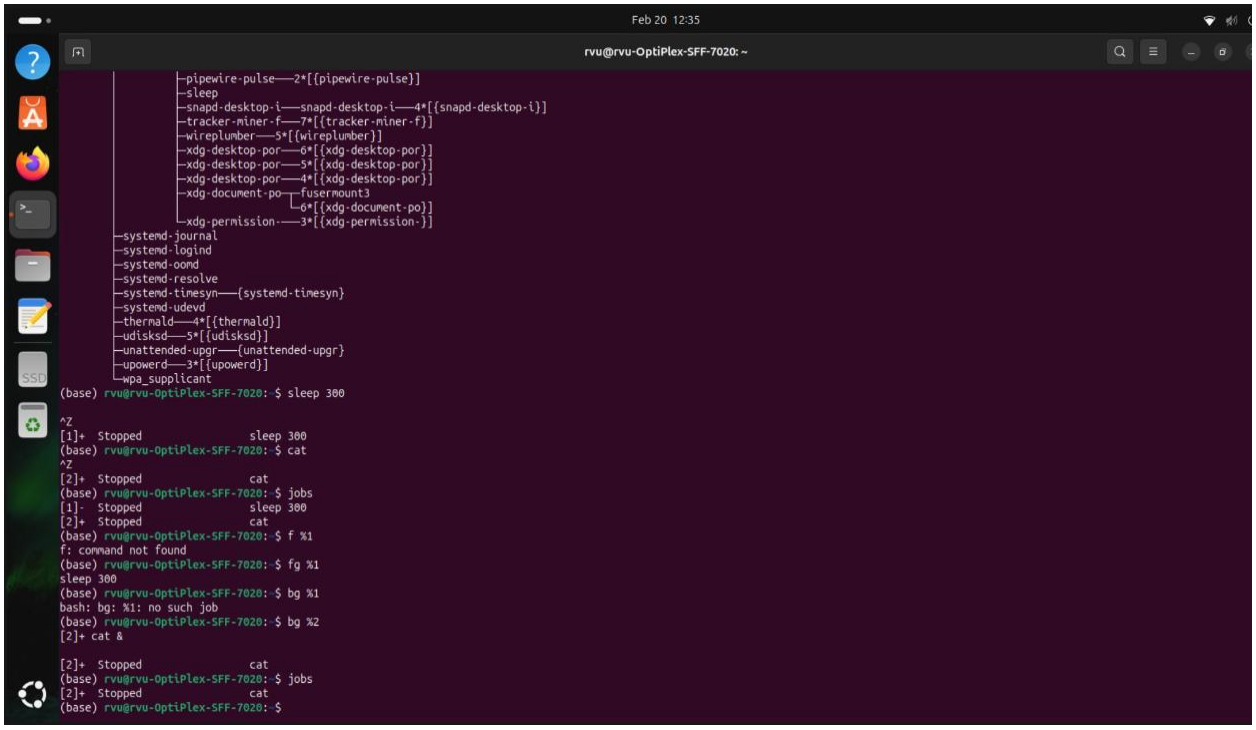
NAME : BHAVYA SAHITHI G

USN: 1RUA25BCA0014

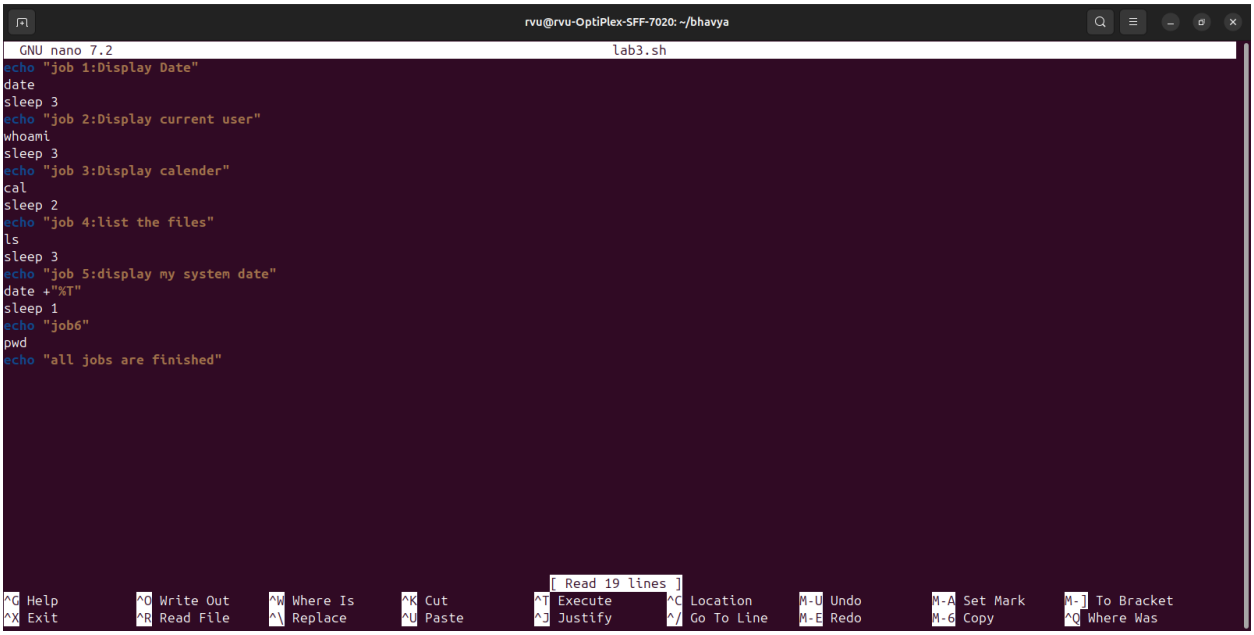
1. **Process Creation in Unix Shell: Creating Child Processes and Displaying Parent-Child Relationships Using ps.**



2. **Process Control: Create a shell script that demonstrates process control commands like bg, fg, jobs, and kill. Implement a simple background task and manage its execution.**



3. **Basic Shell Script for Job Scheduling: Develop a shell script that takes a list of commands and executes them with a specified delay using the sleep command.Demonstrate job scheduling in the shell.**



Output

```
rvu@rvu-OptiPlex-SFF-7020:~/bhavya$ nano lab3.sh
rvu@rvu-OptiPlex-SFF-7020:~/bhavya$ chmod +x lab3.sh
rvu@rvu-OptiPlex-SFF-7020:~/bhavya$ ./lab3.sh
job 1:Display Date
Fri Feb 27 11:48:16 AM IST 2026
job 2:Display current user
rvu
job 3:Display calender
   February 2026
Su Mo Tu We Th Fr Sa
 1  2  3  4  5  6  7
 8  9 10 11 12 13 14
15 16 17 18 19 20 21
22 23 24 25 26 27 28


job 4:list the files
lab3.sh
job 5:display my system date
11:48:27
job6
/home/rvu/bhavya
all jobs are finished
rvu@rvu-OptiPlex-SFF-7020:~/bhavya$
```

4. **File Operations: Create a shell script that performs file operations, such as creating, copying, moving, and deleting files using commands like touch, cp, mv, and rm. Include error handling to manage file operations gracefully. In shell scripting (file operations), -f and -e are file test operators used with if**

   <u>CONDITION</u>

   - **-f → Checks if file exists AND is a regular file - if [-f filename] ⍰ normal file**
   - **-e → Checks if file exists (any type) - if [-e filename] ⍰ File exists (file / directory / link anything)**

```
  GNU nano 7.2                                                    prog4.sh
#!/bin/bash

while true
do
echo "-------------- FILE OPERATIONS ----------"
echo "1. Create File"
echo "2. Copy File"
echo "3. Move File"
echo "4. Delete File"
echo "5. Display File"
echo "6. Exit"

echo -n "Enter your choice: "
read choice

if [ $choice -eq 1 ]; then
    read -p "Enter filename to create: " fname
    if [ -f "$fname" ]; then
        echo "Error: File already exists!"
    else
        read -p "Enter content: " content
        echo "$content" > "$fname"
        echo "File '$fname' created successfully."
    fi

elif [ $choice -eq 2 ]; then
    read -p "Enter source filename: " source
    read -p "Enter destination filename: " dest
    if [ -f "$source" ]; then
        cp "$source" "$dest"
        echo "File copied successfully."
    else
        echo "Error: Source file does not exist!"
    fi

elif [ $choice -eq 3 ]; then
```

```
    if [ -f "$source" ]; then
        mv "$source" "$dest"
        echo "File moved successfully."
    else
        echo "Error: File does not exist!"
    fi

elif [ $choice -eq 4 ]; then
    read -p "Enter filename to delete: " fname
    if [ -f "$fname" ]; then
        rm "$fname"
        echo "File deleted successfully."
    else
        echo "Error: File does not exist!"
    fi

elif [ $choice -eq 5 ]; then
    read -p "Enter filename to display: " fname
    if [ -f "$fname" ]; then
        echo "Content of '$fname':"
        cat "$fname"
    else
        echo "Error: File does not exist!"
    fi

elif [ $choice -eq 6 ]; then
    echo "Exiting program..."
    break

else
    echo "Invalid choice!"
fi

echo ""
done
```

OUTPUT

```
bhavya@BHAVYA:~/OS Lab$ ./prog4.sh
-------------- FILE OPERATIONS ----------
1. Create File
2. Copy File
3. Move File
4. Delete File
5. Display File
6. Exit
Enter your choice: 1
Enter filename to create: file4.txt
Enter content: Hello World!
File 'file4.txt' created successfully.

-------------- FILE OPERATIONS ----------
1. Create File
2. Copy File
3. Move File
4. Delete File
5. Display File
6. Exit
Enter your choice: 5
Enter filename to display: file4.txt
Content of 'file4.txt':
Hello World!

-------------- FILE OPERATIONS ----------
1. Create File
2. Copy File
3. Move File
4. Delete File
5. Display File
6. Exit
Enter your choice: 2
Enter source filename: file4.txt
Enter destination filename: file5.txt
File copied successfully.
```

```
-------------- FILE OPERATIONS ----------
1. Create File
2. Copy File
3. Move File
4. Delete File
5. Display File
6. Exit
Enter your choice: 3
Enter filename to move: file5.txt
Enter new filename: newfile.txt
File moved successfully.

-------------- FILE OPERATIONS ----------
1. Create File
2. Copy File
3. Move File
4. Delete File
5. Display File
6. Exit
Enter your choice: 4
Enter filename to delete: file4.txt
File deleted successfully.

-------------- FILE OPERATIONS ----------
1. Create File
2. Copy File
3. Move File
4. Delete File
5. Display File
6. Exit
Enter your choice: 6
Exiting program...
```

5. **File Permissions: Write a shell script to modify file permissions and ownershig using chmod and chown. Include user input for the filename and desired permissions and display the new file permissions.**

```
  GNU nano 7.2                                                    prog5.sh
echo "---- File Permission & Ownership ----"
echo "Enter file name:"
read file
if [ ! -f "$file" ]; then
echo "File not found!"
exit
fi
echo "Enter permission (e.g., 755):"
read perm
chmod $perm $file
echo "Enter new owner name:"
read owner
chown $owner $file
echo "Updated file details:"
ls -l $file
```

OUTPUT

```
bhavya@BHAVYA:~/OS Lab$ chmod +x prog5.sh
bhavya@BHAVYA:~/OS Lab$ ./prog5.sh
---- File Permission & Ownership ----
Enter file name:
Bhavya
File not found!
```

6. **Directory Management: Develop a shell script that creates, lists, and deletes directories. Use commands like mkdir, ls, and rmdir, and include user prompts for directory names.**

```
echo "Lab 6: Directory Management"
while true
do
    echo ""
    echo "1. Create Directory"
    echo "2. List Directories"
    echo "3. Delete Directory"
    echo "4. Exit"
    echo "Enter your choice:"
    read choice
    case $choice in
    1)
        echo "Enter directory name to create:"
        read dirname
        mkdir "$dirname"
        if [ $? -eq 0 ]; then
            echo "Directory '$dirname' created successfully."
        else
            echo "Error creating directory."
        fi
        ;;
    2)
        echo "Directories in current location:"
        ls -d */
        ;;
    3)
        echo "Enter directory name to delete:"
        read dirname
        rmdir "$dirname"
        if [ $? -eq 0 ]; then
            echo "Directory '$dirname' deleted successfully."
        else
            echo "Error deleting directory (may not be empty)."
        fi
        ;;
    4)
```

```
        fi
        ;;
    4)
        echo "Exiting program..."
        break
        ;;
    *)
        echo "Invalid choice. Try again!"
        ;;
    esac
done
```

OUTPUT

```
bhavya@BHAVYA:~/OS Lab$ ./lab6.sh
Lab 6: Directory Management

1. Create Directory
2. List Directories
3. Delete Directory
4. Exit
Enter your choice:
1
Enter directory name to create:
file6.txt
Directory 'file6.txt' created successfully.

1. Create Directory
2. List Directories
3. Delete Directory
4. Exit
Enter your choice:
2
Directories in current location:
file6.txt/

1. Create Directory
2. List Directories
3. Delete Directory
4. Exit
Enter your choice:
3
Enter directory name to delete:
file6.txt
Directory 'file6.txt' deleted successfully.

1. Create Directory
2. List Directories
3. Delete Directory
4. Exit
Enter your choice:
4
Exiting program...
```

7. Dynamic Memory Check: Write a shell script that monitors memory usage on the system using commands like free and top. Display the current memory usage and available memory.

Memory usage → shows how much RAM is being used by the system

Free memory → available RAM for new processes
free command → gives summary of total, used, and free memory
top command → shows real-time system usage (CPU + memory)

-h option in free → displays values in human-readable format (MB/GB)

```
  GNU nano 7.2                                                    lab7.sh
echo "Lab 7: Memory Usage Monitor"
echo ""
echo "Current Memory Status (using free command):"
free -h
echo ""
echo "Detailed Memory Info (using top command):"
top -b -n 1 | grep "MiB Mem"
echo ""
echo "Available Memory:"
free -h | grep "Mem" | awk '{print "Available: " $7}'
```

OUTPUT

```
bhavya@BHAVYA:~/OS Lab$ nano lab7.sh
bhavya@BHAVYA:~/OS Lab$ chmod +x lab7.sh
bhavya@BHAVYA:~/OS Lab$ ./lab7.sh
Lab 7: Memory Usage Monitor

Current Memory Status (using free command):
               total        used        free      shared  buff/cache   available
Mem:           7.5Gi       522Mi       7.0Gi       3.6Mi       154Mi       7.0Gi
Swap:          2.0Gi          0B       2.0Gi

Detailed Memory Info (using top command):
MiB Mem :   7670.7 total,   7138.9 free,    525.8 used,    155.0 buff/cache

Available Memory:
Available: 7.0Gi
```

8. Environment Variables: Create a shell script that sets, exports, and displays environment variables. Use printenv and env to show the current environment settings.

```bash
#!/bin/bash

echo "Lab 8: Environment Variables Demo"

# Setting variable
name="BHAVYA SAHITHI G"
course="Operating Systems"

echo ""
echo "Local Variables:"
echo "Name: $name"
echo "Course: $course"

# Exporting variables
export name
export course

echo ""
echo "Using printenv command:"
printenv name
printenv course

echo ""
echo "All Environment Variables (using env):"
env
```

OUTPUT

```
PWD=/home/bhavya/OS Lab
LOGNAME=bhavya
HOME=/home/bhavya
LANG=C.UTF-8
WSL_INTEROP=/run/WSL/296_interop
LS_COLORS=rs=0:di=01;34:ln=01;36:mh=00:pi=40;33:so=01;35:do=01;35:bd=40;33;01:cd=40;33;01:or=40;31;01:mi=00:su=37;41:sg=30;43:ca=00:tw=30;42:ow=34;42:st=37;
44:ex=01;32:*.tar=01;31:*.tgz=01;31:*.arc=01;31:*.arj=01;31:*.taz=01;31:*.lha=01;31:*.lz4=01;31:*.lzh=01;31:*.lzma=01;31:*.tlz=01;31:*.txz=01;31:*.tzo=01;31
:*.t7z=01;31:*.zip=01;31:*.z=01;31:*.dz=01;31:*.gz=01;31:*.lrz=01;31:*.lz=01;31:*.lzo=01;31:*.xz=01;31:*.zst=01;31:*.tzst=01;31:*.bz2=01;31:*.bz=01;31:*.tbz
=01;31:*.tbz2=01;31:*.tz=01;31:*.deb=01;31:*.rpm=01;31:*.jar=01;31:*.war=01;31:*.ear=01;31:*.sar=01;31:*.rar=01;31:*.alz=01;31:*.ace=01;31:*.zoo=01;31:*.cpi
o=01;31:*.7z=01;31:*.rz=01;31:*.cab=01;31:*.wim=01;31:*.swm=01;31:*.dwm=01;31:*.esd=01;31:*.avif=01;35:*.jpg=01;35:*.jpeg=01;35:*.mjpg=01;35:*.mjpeg=01;35:*
.gif=01;35:*.bmp=01;35:*.pbm=01;35:*.pgm=01;35:*.ppm=01;35:*.tga=01;35:*.xbm=01;35:*.xpm=01;35:*.tif=01;35:*.tiff=01;35:*.png=01;35:*.svg=01;35:*.svgz=01;35
:*.mng=01;35:*.pcx=01;35:*.mov=01;35:*.mpg=01;35:*.mpeg=01;35:*.m2v=01;35:*.mkv=01;35:*.webm=01;35:*.webp=01;35:*.ogm=01;35:*.mp4=01;35:*.m4v=01;35:*.mp4v=0
1;35:*.vob=01;35:*.qt=01;35:*.nuv=01;35:*.wmv=01;35:*.asf=01;35:*.rm=01;35:*.rmvb=01;35:*.flc=01;35:*.avi=01;35:*.fli=01;35:*.flv=01;35:*.gl=01;35:*.dl=01;3
5:*.xcf=01;35:*.xwd=01;35:*.yuv=01;35:*.cgm=01;35:*.emf=01;35:*.ogv=01;35:*.ogx=01;35:*.aac=00;36:*.au=00;36:*.flac=00;36:*.m4a=00;36:*.mid=00;36:*.midi=00;
36:*.mka=00;36:*.mp3=00;36:*.mpc=00;36:*.ogg=00;36:*.ra=00;36:*.wav=00;36:*.oga=00;36:*.opus=00;36:*.spx=00;36:*.xspf=00;36:*~=00;90:*#=00;90:*.bak=00;90:*.
crdownload=00;90:*.dpkg-dist=00;90:*.dpkg-new=00;90:*.dpkg-old=00;90:*.dpkg-tmp=00;90:*.old=00;90:*.orig=00;90:*.part=00;90:*.rej=00;90:*.rpmnew=00;90:*.rpm
orig=00;90:*.rpmsave=00;90:*.swp=00;90:*.tmp=00;90:*.ucf-dist=00;90:*.ucf-new=00;90:*.ucf-old=00;90:
WAYLAND_DISPLAY=wayland-0
LESSCLOSE=/usr/bin/lesspipe %s %s
TERM=xterm-256color
LESSOPEN=| /usr/bin/lesspipe %s
USER=bhavya
DISPLAY=:0
SHLVL=2
XDG_RUNTIME_DIR=/run/user/1000/
WSLENV=WT_SESSION:WT_PROFILE_ID:
XDG_DATA_DIRS=/usr/local/share:/usr/share:/var/lib/snapd/desktop
PATH=/usr/local/sbin:/usr/local/bin:/usr/sbin:/usr/bin:/sbin:/bin:/usr/games:/usr/local/games:/usr/lib/wsl/lib:/mnt/c/Windows/system32:/mnt/c/Windows:/mnt/c
/Windows/System32/Wbem:/mnt/c/Windows/System32/WindowsPowerShell/v1.0/:/mnt/c/Windows/System32/OpenSSH/:/mnt/c/Program Files/dotnet/:/mnt/c/Program Files/Gi
t/cmd:/mnt/c/Users/Bhavy/AppData/Local/Programs/Python/Python314/Scripts/:/mnt/c/Users/Bhavy/AppData/Local/Programs/Python/Python314/:/mnt/c/Users/Bhavy/App
Data/Local/Programs/Python/Launcher/:/mnt/c/Users/Bhavy/AppData/Local/Microsoft/WindowsApps:/mnt/c/Users/Bhavy/AppData/Local/Python/bin:/snap/bin
DBUS_SESSION_BUS_ADDRESS=unix:path=/run/user/1000/bus
HOSTTYPE=x86_64
course=Operating Systems
PULSE_SERVER=unix:/mnt/wslg/PulseServer
WT_PROFILE_ID={e009a919-0408-5508-b37f-b75ca4f2cac0}
OLDPWD=/home/bhavya
name=BHAVYA SAHITHI G
_=/usr/bin/env
```

9. I/O Redirection in Unix Shell: Manipulating Data Streams with Enhanced File Operations.

```
  GNU nano 7.2                                                    lab9.sh
#!/bin/bash

echo "Lab 9: I/O Redirection Demo"

# Output redirection (overwrite)
echo "Hello, this is OS Lab" > file1.txt
echo "Content written to file1.txt"

# Append output
echo "Appending new line" >> file1.txt
echo "Content appended to file1.txt"

# Input redirection
echo ""
echo "Reading from file1.txt using input redirection:"
cat < file1.txt

# Error redirection
echo ""
echo "Trying to read non-existing file:"
cat file_not_exist.txt 2> error.txt
echo "Error message stored in error.txt"

# Display error file
echo ""
echo "Error file content:"
cat error.txt
```

OUTPUT

```
bhavya@BHAVYA:~/OS Lab$ nano lab9.sh
bhavya@BHAVYA:~/OS Lab$ chmod +x lab9.sh
bhavya@BHAVYA:~/OS Lab$ ./lab9.sh
Lab 9: I/O Redirection Demo
Content written to file1.txt
Content appended to file1.txt

Reading from file1.txt using input redirection:
Hello, this is OS Lab
Appending new line

Trying to read non-existing file:
Error message stored in error.txt

Error file content:
cat: file_not_exist.txt: No such file or directory
bhavya@BHAVYA:~/OS Lab$
```

10. Command Piping in Unix Shell: Exploring Operating System Features.

```
#!/bin/bash

echo "Command Piping Demo"

echo -e "apple\nbanana\napple\norange\nbanana\napple" > fruits.txt

echo "Sample file 'fruits.txt' created with content:"
cat fruits.txt
echo

total_lines=$(cat fruits.txt | wc -l)
echo "Total lines in fruits.txt: $total_lines"

apple_count=$(cat fruits.txt | grep "apple" | wc -l)
echo "Number of times 'apple' appears: $apple_count"

echo "Sorted unique fruits:"
cat fruits.txt | sort | uniq
echo

echo "Lines containing 'banana':"
cat fruits.txt | grep "banana"

rm fruits.txt
echo
echo "Demo complete. Sample file removed."
```

OUTPUT

```
bhavya@BHAVYA:~/OS Lab$ nano lab10.sh
bhavya@BHAVYA:~/OS Lab$ chmod +x lab10.sh
bhavya@BHAVYA:~/OS Lab$ ./lab10.sh
Command Piping Demo
Sample file 'fruits.txt' created with content:
apple
banana
apple
orange
banana
apple

Total lines in fruits.txt: 6
Number of times 'apple' appears: 3
Sorted unique fruits:
apple
banana
orange

Lines containing 'banana':
banana
banana

Demo complete. Sample file removed.
bhavya@BHAVYA:~/OS Lab$
```